

Support-Need Index from Gaze and Hand Signals

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Technical Assignment Task 2

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1 Problem formulation

For Part B, I defined a simple *Support-Need Index* (SNI) to estimate moments where the operator may need support during the assembly task. The idea is that support may be useful when the operator pauses, searches visually, or checks the instruction sheet. This is relevant for an XR guidance system because help should ideally appear when it is needed, rather than being shown continuously.

There is no direct ground-truth label for ‘needs support’. Therefore, I used instruction-consulting behaviour as a weak proxy. Frames where the operator checks or browses instructions are treated as approximate support-seeking moments. This proxy is imperfect, since some instruction checks are routine.

2 Method

The SNI combines three interpretable signals from gaze and hand data:

1. **Gaze entropy**: high when gaze is spread over many image regions, suggesting visual search.
2. **Hand stall**: high when hand-joint motion slows down, suggesting a pause or hesitation.
3. **Instruction-AOI dwell**: high when gaze stays inside the instruction-sheet region.

All signals are computed over a 30-frame rolling window. Each signal is then z-scored per recording to normalise for different operator baselines. The three z-scored signals are averaged and passed through a sigmoid:

$$\text{SNI}_t = \sigma \left(\frac{z_{\text{gaze},t} + z_{\text{hand},t} + z_{\text{AOI},t}}{3} \right). \quad (1)$$

This gives a value in $[0, 1]$, where a higher value indicates stronger evidence of possible support need. I used equal weights because the dataset is small and I did not want to overfit a learned fusion model.

2.1 Gaze entropy

The image is divided into an 8×8 grid. For each 30-frame window, valid gaze points are counted in the grid cells, producing a spatial gaze distribution. Shannon entropy is then computed as:

$$H_t = - \sum_i p_{i,t} \log_2(p_{i,t}), \quad (2)$$

where $p_{i,t}$ is the fraction of gaze samples in tile i . Low entropy means gaze is focused; high entropy means gaze is scattered.

2.2 Hand stall

Hand stall is based on hand-joint motion. Instead of using only the hand centroid, the implementation uses joint-level movement so that small finger or wrist motion during manipulation is not missed. The motion signal is smoothed over 30 frames and inverted, so low movement becomes high stall evidence.

2.3 Instruction-AOI dwell

The instruction sheet is approximated by a fixed lower-left image region:

$$x < 0.35W \quad \text{and} \quad y > 0.25H. \quad (3)$$

For a 1280×720 frame, this is approximately:

$$x < 448, \quad y > 180. \quad (4)$$

The AOI dwell signal is the fraction of valid gaze samples inside this region within the 30-frame window.

3 Episode threshold

To convert the continuous SNI curve into binary support-need episodes, I used a per-recording adaptive threshold:

$$\theta_r = \mu_{\text{SNI},r} + \sigma_{\text{SNI},r}. \quad (5)$$

A frame is marked as a support-need episode when:

$$\text{SNI}_t > \theta_r. \quad (6)$$

4 Evaluation

I evaluated SNI against the weak instruction-consulting proxy ROC-AUC.

The per-recording AUC values are shown below.

Table 1: Per-recording AUC values.

Recording	AUC
22_assy_0_1	0.65
22_assy_2_3	0.30
25_assy_0_1	0.70
25_assy_2_1	0.79
27_assy_0_1	0.80
Pooled	0.72

The below-chance result for 22_assy_2_3 suggests that the proxy and the index disagree for that recording.

5 Ablation study

I also evaluated each individual signal. AOI dwell performs best because it is very close to the instruction-consulting proxy itself.

Table 2: Ablation study using pooled AUC.

Signal	Pooled AUC
Instruction-AOI dwell only	0.78
Fused SNI	0.72
Gaze entropy only	0.61
Hand stall only	0.41

Although AOI dwell has the highest AUC, the fused SNI is still useful because it is not only an instruction-checking detector. It can also respond to other hesitation cues, such as scattered gaze or stalled hand motion.

6 Limitations

- The main limitation is that instruction consulting is a weak proxy label for support need.
- The dataset is also very small, so per-recording AUC values have high variance.
- The fusion weights and threshold are fixed rather than learned
- The instruction AOI is a simple global rectangle that may need adjustment for other camera views.

7 practical relevance

In an XR guidance system, SNI could trigger extra support for the current procedure step.